

Chanu Kalubowila

✉ chanuk@uom.lk | 📞 +94 70 197 6460 | 🌐 github.com/CK20000 | 🌐 ck20000.github.io

EDUCATION

Bachelor of the Science of Engineering (Hons.) Aug. 2021 – Jul. 2025
University of Moratuwa, Sri Lanka (Washington Accord Accredited)

- Mechatronics Systems Engineering Stream; Minor in Mathematics
- **GPA: 3.79/4.0** — Ranked **9th** in Mechanical Engineering (120 students)
- Dean's List: 4 semesters (GPA > 3.80)

GCE Advanced Level Examination Jan. 2017 – Aug. 2019
Ananda College, Colombo, Sri Lanka — 3A Passes, Z-Score: 2.4313 (Top 1% of 47,000+)

TEACHING & INDUSTRIAL EXPERIENCE

Temporary Lecturer May 2026 – Present
Department of Mechanical Engineering, University of Moratuwa

- Lectures, labs, and workshops: PCB Designing, Instrumentation System Design, Industrial Automation, Control Systems, Mechatronic System Design Project, and MEMS.

Academic Instructor Nov. 2025 – May 2026
Department of Mechanical Engineering, University of Moratuwa

Trainee Mechatronics Engineer Dec. 2023 – Jun. 2024
Colombo International Container Terminals, Sri Lanka

PUBLICATIONS & PATENTS

Refereed Conference Papers

1. **Kalubowila K.D.C.S.**, Gayan K.A.K., Perera M.A.Y., Jayathilaka W.A.D.M., Amarasinghe Y.W.R. "Passive Mechanical Retention of Capillary Tubes for Machine Vision Compatible High Speed PCV Test Centrifugation Rotor." In: *Moratuwa Engineering Research Conference (MERCon)*, Sri Lanka, Aug. 2026. **Best Paper Award**. DOI: [10.1109/MERCon71835.2026.11691396](https://doi.org/10.1109/MERCon71835.2026.11691396)
2. Perera M.A.Y., **Kalubowila K.D.C.S.**, Gayan K.A.K., Amarasinghe Y.W.R., Jayathilaka W.A.D.M. "Design and Manufacturing Optimization of a Highspeed PCV Centrifuge Machine." In: *MERCon*, Sri Lanka, Aug. 2025. DOI: [10.1109/MERCon67903.2025.11217024](https://doi.org/10.1109/MERCon67903.2025.11217024)

Journal Papers (Under Review)

3. **Kalubowila K.D.C.S.**, et al. "Design and Clinical Validation of a Vision-Based Automated Centrifuge Instrument for PCV Measurement." *Measurement* (Elsevier).
4. Perera M.A.Y., **Kalubowila K.D.C.S.**, et al. "Automated Hematocrit Estimation from Multi-Tube Centrifuge Rotor Images via Oriented Detection and Axis-Constrained Boundary Regression." *Biomedical Signal Processing and Control* (Elsevier).
5. Gayan K.A.K., **Kalubowila K.D.C.S.**, Perera M.A.Y., Amarasinghe Y.W.R., Jayathilaka W.A.D.M. "Optimal PID Controller Design for Brushed DC Motors using CMA-ES Optimization." *Engineering Research Express* (IOP).

Patents & Registered Designs

- **Patent (Pending)**: "A Rotor Disk in Centrifuge Machines for the Packed Cell Volume Test." App. No. **P/1/023851**, National Intellectual Property Office of Sri Lanka (NIPO).
- **Industrial Design (Registered)**: "Centrifuge Machine." App. No. **14341**, University of Moratuwa.
- **Industrial Designs (Filed)**: STM32H743 Flight Controller Carrier Board; Bare-Metal Sensorless FOC ESC Board.

HONOURS & AWARDS

Best Paper Award — MERCon 2026 (Mechanical Systems Track; Final Year Project) 2026
IESL Undergraduate Inventor of the Year 2025 — 1st Place (Institution of Engineers, Sri Lanka; Final Year Project) Jul. 2025
Mobile Award — Best Mechanical Engineering Final Year Project (McLarens Lubricants) 2025
Dean's List — Faculty of Engineering, University of Moratuwa (4 semesters, GPA > 3.80) 2021–2025
Baseball — University Colors & Inter-University Champions (pitcher, shortstop) 2023, 2024

RESEARCH & ENGINEERING PROJECTS

Vision-Based PCV Analyzer — Final Year Research Project Aug. 2021 – Jul. 2025

University of Moratuwa •  [Portfolio](#)

Novel lidless passively-retaining centrifuge rotor + deep-learning segmentation pipeline for automated hematocrit measurement.

Flapping-Wing Robot — Full-Stack Design, Simulation & Prototype Aug. 2023 – Dec. 2024

Foundation of Cross-Domain Simulation Work •  [Portfolio](#)

Fabricated a 110 cm, 4.3 N bio-inspired ornithopter from scratch and built custom ROS+Gazebo aerodynamic plugins as an open-source alternative to MATLAB/Simulink in flapping-wing research.

Bare-Metal Sensorless FOC ESC Kernel May 2026 – Present

STM32G431 •  [Portfolio](#)

Wrote a sensorless FOC kernel for low-inductance UAV motors achieving a 2,728-cycle (16.05 μ s) control loop at 24 kHz with a State-Space Flux Observer and hardware CORDIC acceleration.

Betaflight Flight Controller — Bare-Metal Port Jan. 2026 – Present

Industrial H743 Devboard •  [Portfolio](#)

Brought Betaflight onto a generic industrial H743 devboard via custom target files and a 4-layer carrier board, fixing three silicon-level bugs that blocked firmware boot.

PCB Digital Twin — Open-Source Plugin Sept. 2026 – Present

Independent Research •  [Portfolio](#)

Built a real-time visualizer that renders a KiCad PCB as a living system (per-net currents, voltages, 2D thermal fields) driven by firmware on a virtual STM32 in Renode.

Lab-on-a-Chip Dielectrophoresis Device (MEMS) Jul. 2024 – Oct. 2024

Design of a Microfluidic Blood Particle Separation LOC

Autonomous Miniature Taxi Robot Mar. 2025 – Jul. 2025

PID Navigation + ML Sign Detection

Fundamental SCADA System Jul. 2024 – Oct. 2024

ESP32 + MySQL + Closed-Loop Thermal Control

IoT-Based Weather Monitoring System 2023

Embedded Sensor Network

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, ROS1, LaTeX

Embedded: STM32 (F4/G4/H7), Renode, PlatformIO, STM32CubeIDE, ARM GCC

Simulation: ngspice, ANSYS, COMSOL, MSC Adams, Gazebo, MATLAB/Simulink

PCB Design: KiCad, Proteus, Gerber generation, BOM management

PLC/HMI: Siemens TIA Portal, Delta, HMI design

Control: PID, FOC, SMO, Active Flux Observer, CORDIC, State-Space Observers

Vision: YOLOv8-ORB, U-Net, PyTorch, OpenCV, FFT analysis

Hardware: Soldering, oscilloscope, logic analyzer, electrical wiring, microcontroller interfacing, arc/MIG welding

EXTRACURRICULAR & VOLUNTEER

Organizing Committee — MERS 2025 (Mechanical Engineering Research Symposium, UoM) Jan. 2026

Organizing Committee — MechatronX 2023 (EXMO 2023, UoM) Jul. 2023

Safety Committee — EXMO 2023 (UoM) Jul. 2023

REFERENCES

Prof. Y. W. R. Amarasinghe

Program Coordinator

BSc Eng Hons in Mechatronics and Automation Systems Eng.

Department of Mechanical Engineering

University of Moratuwa, Sri Lanka

ranama@uom.lk

Dr. W. A. D. M. Jayathilaka

Stream Coordinator

Mechatronics and Automation Systems Eng.

Department of Mechanical Engineering

University of Moratuwa, Sri Lanka

dumithj@uom.lk